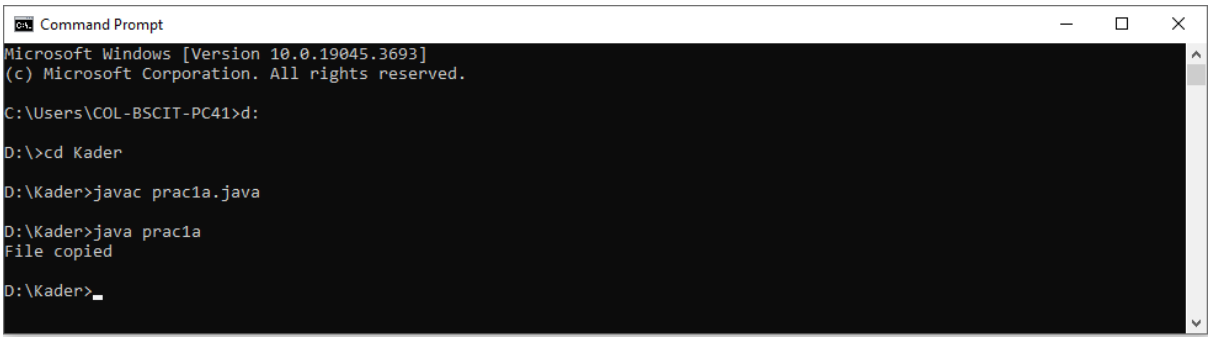
KISHINCHAND CHELLARAM COLLEGE, MUMBAI - 20
T.Y. B.Sc. I.T. Semester VI

Practical 1A

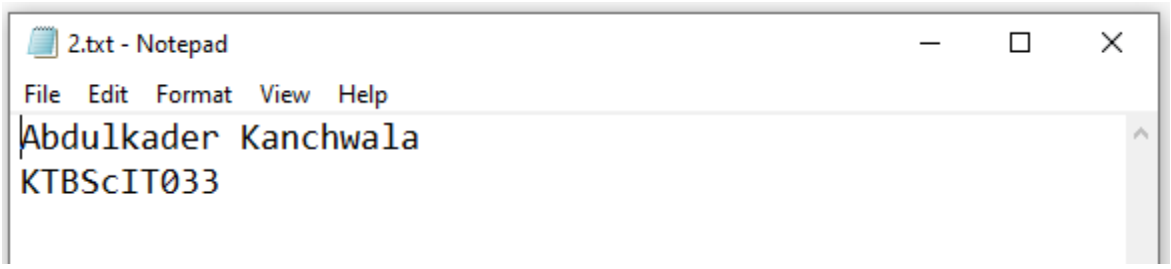
Code:

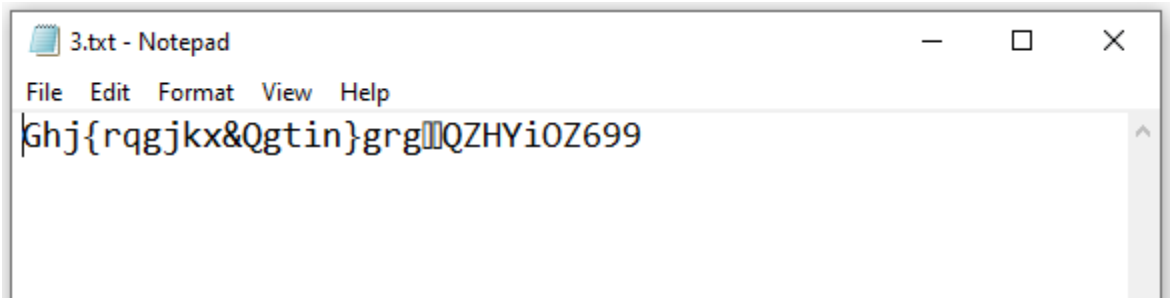
```
import java.io.*;
class prac1a {
   public static void main(String args[]) {
      int i;
      try {
         File f1 = new File("1.txt");
         File f2 = new File("2.txt");
         File f3 = new File("3.txt");
         FileInputStream fis = new FileInputStream(f1);
         FileOutputStream fos = new FileOutputStream(f2);
         FileOutputStream os = new FileOutputStream(f3);
         while ((i = fis.read()) != -1) {
            fos.write(i);
            os.write(i + 6);
         }
         System.out.println("File copied");
      } catch (Exception e) {
         e.printStackTrace();
      }
   }
}
```

Output:


Command Prompt
Microsoft Windows [Version 10.0.19045.3693]
(c) Microsoft Corporation. All rights reserved.
C:\Users\COL-BSCIT-PC41>d:
D:\>cd Kader
D:\Kader>javac prac1a.java
D:\Kader>java prac1a
File copied
D:\Kader>

1.txt - Notepad
File Edit Format View Help
Abdulkader Kanchwala
KTBScIT033


2.txt - Notepad
File Edit Format View Help
Abdulkader Kanchwala
KTBScIT033


3.txt - Notepad
File Edit Format View Help
Ghj{rqgjkx&Qgtin}grg QZHYiOZ699

Practical 1B

Code:

```
import java.io.*;
class prac1b {
   public String Encrypt(int shift, String line) {
      String result = "";
      int offset;
      for (int i = 0; i < line.length(); i++) {
         offset = ((int) line.charAt(i) + shift) % 256;
         result += (char) (offset);
      }
      return result;
   }

   public String Decrypt(int shift, String line) {
      String result = "";
      int offset;
      for (int i = 0; i < line.length(); i++) {
         offset = ((int) line.charAt(i) - shift) % 256;
         if (offset < 0) {
            offset += 256;
         }
         result += (char) (offset);
      }
      return result;
   }

   public static void main(String args[])
         throws IOException {
      prac1b b = new prac1b();
      BufferedReader br = new BufferedReader(new InputStreamReader(System.in));
      int choice;
      System.out.println("Menu:\n1: Encryption\n2:Decryption");
      choice = Integer.parseInt(br.readLine());
      System.out.println("Enter the shift");
      int shift = Integer.parseInt(br.readLine());
      System.out.println("Enter the line");
      String line = br.readLine();
      System.out.println("Result: ");
      switch (choice) {
         case 1: {
            System.out.println(b.Encrypt(shift, line));
            break;
```

```
        }
        case 2: {
            System.out.println(b.Decrypt(shift, line));
            break;
        }
        default: {
            System.out.println("Invalid choice");
        }
    }
  }
}
```

Output:

Practical 1C

Code:

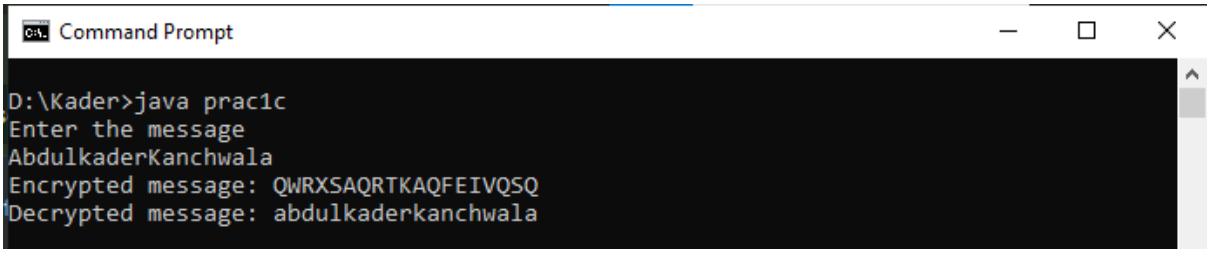
```
import java.util.Scanner;
public class prac1c {
   public static char p[] = { 'a', 'b', 'c', 'd', 'e', 'f', 'g', 'h', 'i', 'j', 'k', 'l', 'm', 'n', 'o', 'p', 'q', 'r',
         's', 't', 'u', 'v', 'w', 'x', 'y', 'z' };
   public static char ch[] = { 'Q', 'W', 'E', 'R', 'T', 'Y', 'U', 'I', 'O', 'P', 'A', 'S', 'D', 'F', 'G', 'H', 'J',
         'K', 'L', 'Z', 'X', 'C', 'V', 'B', 'N', 'M' };

   public static String doEncryption(String s) {
      char c[] = new char[(s.length())];
      for (int i = 0; i < s.length(); i++) {
         for (int j = 0; j < 26; j++) {
            if (p[j] == s.charAt(i)) {
               c[i] = ch[j];
               break;
            }
         }
      }
      return (new String(c));
   }

   public static String doDecryption(String s) {
      char p1[] = new char[(s.length())];
      for (int i = 0; i < s.length(); i++) {
         for (int j = 0; j < 26; j++) {
            if (ch[j] == s.charAt(i)) {
               p1[i] = p[j];
               break;
            }
         }
      }
      return (new String(p1));
   }
```

```
public static void main(String args[]) {
    Scanner sc = new Scanner(System.in);
    System.out.println("Enter the message");
    String en = doEncryption(sc.next().toLowerCase());
    System.out.println("Encrypted message: " + en);
    System.out.println("Decrypted message: " +
        doDecryption(en));
    sc.close();
  }
}
```

Output:



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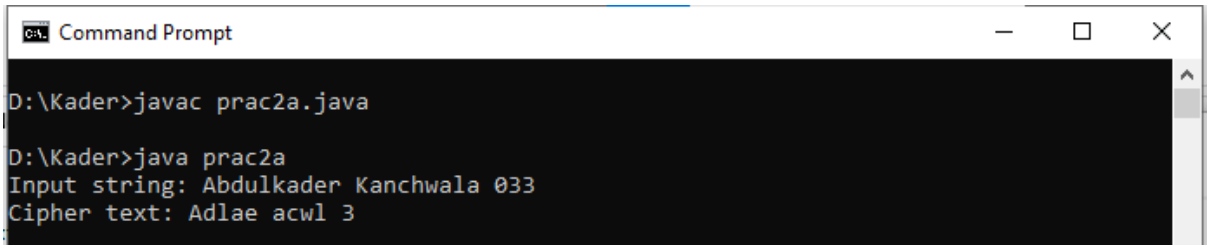
Practical 2A

Code:

```
import java.*;
public class prac2a {
   public static void main(String args[]) {
      String input = "Abdulkader Kanchwala 033";
      String output = "";
      int len = input.length();
      System.out.println("Input string: " + input);
      for (int i = 0; i < len; i += 2) {
         output += input.charAt(i);
      }
      System.out.println("Cipher text: " + output);
   }
}
```

Output:



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Practical 2B

Code:

```
import java.lang.Math;
public class prac2b {
   public static void main(String args[]) {
      String str = new String("Abdulkader");
      char[] arText = str.toCharArray();
      String cipher = new String("KANCHWALAA");
      char[] arCipher = cipher.toCharArray();
      char[] encoded = new char[10];
      System.out.println("Encoded to be..");
      for (int i = 0; i < arText.length; i++) {
         encoded[i] = (char) (arText[i] ^ arCipher[i]);
         System.out.println(encoded[i]);
      }
      System.out.println("\nDecoded to be..");
      for (int i = 0; i < encoded.length; i++) {
         char tem = (char) (encoded[i] ^ arCipher[i]);
         System.out.println(tem);
      }
   }
}
```

Output:

```
D:\Kader>javac prac2b.java

D:\Kader>java prac2b
Encoded to be..


#
*
6
$
<

(
$
3

Decoded to be..
A
b
d
u
l
k
a
d
e
r
```

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Practical 3

Code:

```
import java.math.BigInteger;
import java.security.KeyFactory;
import java.security.KeyPair;
import java.util.*;
import java.math.BigInteger.*;
import java.math.*;

class prac3 {
   public static void main(String args[]) {
      BigInteger a, b, k1, k2, x, y, g, n;
      Scanner s = new Scanner(System.in);
      System.out.println("Enter A's prime number");
      n = s.nextBigInteger();
      System.out.println("Enter B's prime number");
      g = s.nextBigInteger();
      System.out.println("Enter A's secret key");
      x = s.nextBigInteger();
      System.out.println("This key is sent to B");
      System.out.println("Enter B's secret key");
      y = s.nextBigInteger();
      System.out.println("This key is sent to A");
      a = g.modPow(x, n);
      b = g.modPow(y, n);
      k1 = b.modPow(x, n);
      k2 = a.modPow(y, n);
      System.out.println("A's key is " + k1);
      System.out.println("B's key is " + k2);
}   }
```

Output:

```
D:\Kader>java prac3
Enter A's prime number
19
Enter B's prime number
23
Enter A's secret key
5
This key is sent to B
Enter B's secret key
9
This key is sent to A
A's key is 1
B's key is 1

D:\Kader>
```

Practical 4

Code:

```
import javax.crypto.*;
import java.io.*;
import java.security.*;
import java.util.Base64;

public class Prac4 {
   Cipher ecipher;
   Cipher dcipher;

   public Prac4(SecretKey key) {
      try {
         ecipher = Cipher.getInstance("DES");
         dcipher = Cipher.getInstance("DES");
         ecipher.init(Cipher.ENCRYPT_MODE, key);
         dcipher.init(Cipher.DECRYPT_MODE, key);
      } catch (NoSuchPaddingException | NoSuchAlgorithmException | InvalidKeyException e) {
         e.printStackTrace();
      }
   }

   public String Encrypt(String str) {
      try {
         byte[] utf8 = str.getBytes("UTF8");
         byte[] enc = ecipher.doFinal(utf8);
         return Base64.getEncoder().encodeToString(enc);
      } catch (BadPaddingException | IllegalBlockSizeException | UnsupportedEncodingException e) {
         e.printStackTrace();
      }
      return null;
   }

   public String Decrypt(String str) {
      try {
         byte[] dec = Base64.getDecoder().decode(str);
         byte[] utf8 = dcipher.doFinal(dec);
         return new String(utf8, "UTF8");
      } catch (BadPaddingException | IllegalBlockSizeException | UnsupportedEncodingException e) {
         e.printStackTrace();
      }

      return null;
```

```
  }

  public static void main(String args[]) {
    System.out.println();
    System.out.println("Encrypting string using DES Algorithm");
    try {
      SecretKey key = KeyGenerator.getInstance("DES").generateKey();
      Prac4 encrypter = new Prac4(key);
      String s = "Abdulkader";
      String encrypted = encrypter.Encrypt(s);
      String decrypted = encrypter.Decrypt(encrypted);
      System.out.println("Original:" + s);
      System.out.println("Encrypt:" + encrypted);
      System.out.println("Decrypt:" + decrypted);
    } catch (NoSuchAlgorithmException e) {
      e.printStackTrace();
    }
  }
}
```

Output:

```
PS C:\Users\abdka\OneDrive\Desktop\JOURNALS\CIS>  & 'C
a\AppData\Roaming\Code\User\workspaceStorage\591a480fb

Encrypting string using DES Algorithm
Original:Abdulkader
Encrypt:d2YU/g8881obebkdh9edxg==
Decrypt:Abdulkader
PS C:\Users\abdka\OneDrive\Desktop\JOURNALS\CIS>
```

Practical 5

Code:

```
import java.security.*;
import javax.crypto.*;
import java.io.*;
import java.util.Base64;

public class Prac5 {
   Cipher ecipher;
   Cipher dcipher;

   public Prac5(SecretKey key) {
      try {
         ecipher = Cipher.getInstance("AES");
         dcipher = Cipher.getInstance("AES");
         ecipher.init(Cipher.ENCRYPT_MODE, key);
         dcipher.init(Cipher.DECRYPT_MODE, key);
      } catch (Exception e) {
         System.out.println(e.getMessage());
      }
   }

   public String Encrypt(String str) {
      try {
         byte[] utf8 = str.getBytes("UTF-8");
         byte[] enc = ecipher.doFinal(utf8);
         return Base64.getEncoder().encodeToString(enc);
      } catch (Exception e) {
         e.printStackTrace();
      }
      return null;
   }

   public String Decrypt(String str) {
      try {
         byte[] dec = Base64.getDecoder().decode(str);
         byte[] utf8 = dcipher.doFinal(dec);
         return new String(utf8,
               "UTF-8");
      } catch (Exception e) {
         System.out.println(e.getMessage());
      }
```

```
        return null;
    }

    public static void main(String[] args) {
        SecretKey key = null;
        try {
            KeyGenerator keyGen = KeyGenerator.getInstance("AES");
            key = keyGen.generateKey();
        } catch (Exception e) {
            e.printStackTrace();
        }
        Prac5 dese = new Prac5(key);
        String o = "Abdulkader";
        String e = dese.Encrypt(o);
        String d = dese.Decrypt(e);
        System.out.println(o);
        System.out.println(e);
        System.out.println(d);
    }
}
```

Output:

```
PS C:\Users\abdka\OneDrive\Desktop\JOURNALS\CIS>
-preview' '-XX:+ShowCodeDetailsInExceptionMessages
a\jdt_ws\CIS_6f716ac3\bin' 'Prac5'
Abdulkader
453WD+++8LG+UlgDDIElgA==
Abdulkader
```

Practical 6

Code:

```
import java.math.*;
import java.security.*;

public class Prac6 {
   BigInteger p, q, n, d, e, ph, t;
   SecureRandom r;

   public Prac6() {
      r = new SecureRandom();
      p = new BigInteger(4, 100, r);
      q = new BigInteger(4, 100, r);
      System.out.println("Prime numbers p and q are " + p.intValue() + ", " + q.intValue());
      n = p.multiply(q);
      ph = (p.subtract(BigInteger.ONE)).multiply(q.subtract(BigInteger.ONE));

      e = new BigInteger("3");
      while (ph.gcd(e).intValue() > 1 || e.compareTo(ph) >= 0) {
         e = e.add(BigInteger.ONE);
      }

      d = e.modInverse(ph);

      System.out.println("Public key is (" + n.intValue() + "," + e.intValue() + ")");
      System.out.println("Private key is (" + n.intValue() + "," + d.intValue() + ")");

      BigInteger msg = new BigInteger("15");
      System.out.println("\nMessage is: " + msg);

      BigInteger enmsg = encrypt(msg, e, n);
      System.out.println("\nEncrypted Message is: " + enmsg.intValue());

      BigInteger demsg = decrypt(enmsg, d, n);
      System.out.println("\nDecrypted Message is: " + demsg.intValue());
   }

   BigInteger encrypt(BigInteger msg, BigInteger e, BigInteger n) {
      return msg.modPow(e, n);
   }

   BigInteger decrypt(BigInteger msg, BigInteger d, BigInteger n) {
      return msg.modPow(d, n);
   }

   public static void main(String args[]) {
```

```
    Prac6 r = new Prac6();
  }
}
```

Output:

```
PS C:\Users\abdka\OneDrive\Desktop\JOURNALS\CIS>
-preview' '-XX:+ShowCodeDetailsInExceptionMessag
a\jdt_ws\CIS_6f716ac3\bin' 'Prac6'
Prime numbers p and q are 13, 13
Public key is (169,5)
Private key is (169,29)

Message is: 15

Encrypted Message is: 58

Decrypted Message is: 119
```

Practical 7

Code:

```
class Prac7 {
   String pt;
   static char cipher[];

   Prac7(String pt, int[] key) {
      this.pt = pt;
      int s[] = new int[255];
      cipher = new char[pt.length()];
      for (int i = 0; i < s.length; i++) {
         s[i] = i;
      }
      int i = 0;
      int j = 0;
      for (int k = 0; k < pt.length(); k++) {
         int modk = (k % key.length);
         int kc = key[modk];
         j = (s[i] + j + kc) % 256 + 1;
         int temp = s[i];
         s[i] = s[j];
         s[j] = temp;
         int sc = (s[i] + s[j]) % 256;
         int ck = s[sc];
         cipher[k] = (char) (ck ^ (int) pt.charAt(k));
         i = i + 1;
      }
   }

   public static void main(String args[]) {
      int k[] = { 1, 2, 3, 4, 5 };
      String original = "Abdulkader";
      System.out.println("Value is: " + original);
      new Prac7(original, k);
      for (int i = 0; i < cipher.length; i++) {
         System.out.println(" " + cipher[i]);
      }
   }
}
```

Output:

```
PS C:\Users\abdka\OneDrive\Desktop\JOURNALS\CIS>
p' 'C:\Users\abdka\AppData\Roaming\Code\User\works
Value is: Abdulkader
 A
 e
 d
 `
 L
 C
 I
 ]
 "
 %
```

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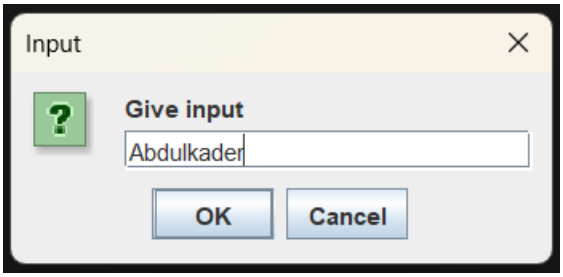
Practical 8

Code:

```
import javax.crypto.Cipher;
import javax.crypto.KeyGenerator;
import javax.crypto.SecretKey;
import javax.swing.JOptionPane;

public class Prac8 {
   public static void main(String args[]) throws Exception {
      KeyGenerator keygen = KeyGenerator.getInstance("Blowfish");
      SecretKey secretkey = keygen.generateKey();
      Cipher cip = Cipher.getInstance("Blowfish");
      cip.init(Cipher.ENCRYPT_MODE, secretkey);
      String inputText = JOptionPane.showInputDialog("Give input");
      byte[] encrypted = cip.doFinal(inputText.getBytes());
      cip.init(Cipher.DECRYPT_MODE, secretkey);
      byte[] decrypted = cip.doFinal(encrypted);
      JOptionPane.showMessageDialog(JOptionPane.getRootFrame(),
            "Encrypted: " + new String(encrypted) + "\n" + "Decrypted:" + new String(decrypted));
      System.exit(0);
   }
}
```

Output: